The Maximum over a Collection of Convex Functions is Convex

Consider

$$V(\theta) = \max_{x \in C} F(x, \theta)$$

where θ lies in a convex subset Θ of $\mathbb{R}^m$ and C is a subset of $\mathbb{R}^n$, $n, m \geq 1$. Let a solution exist for every $\theta \in \Theta$.

Theorem If $F(x, \cdot)$ is convex for every $x \in C$, then $V(\cdot)$ is convex.

Proof: Exercise. (Hint: Let $\theta', \theta'' \in \Theta$, $\lambda \in [0, 1]$, and let x^λ solve the problem at $\theta = \lambda\theta'' + (1-\lambda)\theta'$. Then $V(\lambda\theta'' + (1-\lambda)\theta') = F(x^\lambda, \lambda\theta'' + (1-\lambda)\theta')$ ...).

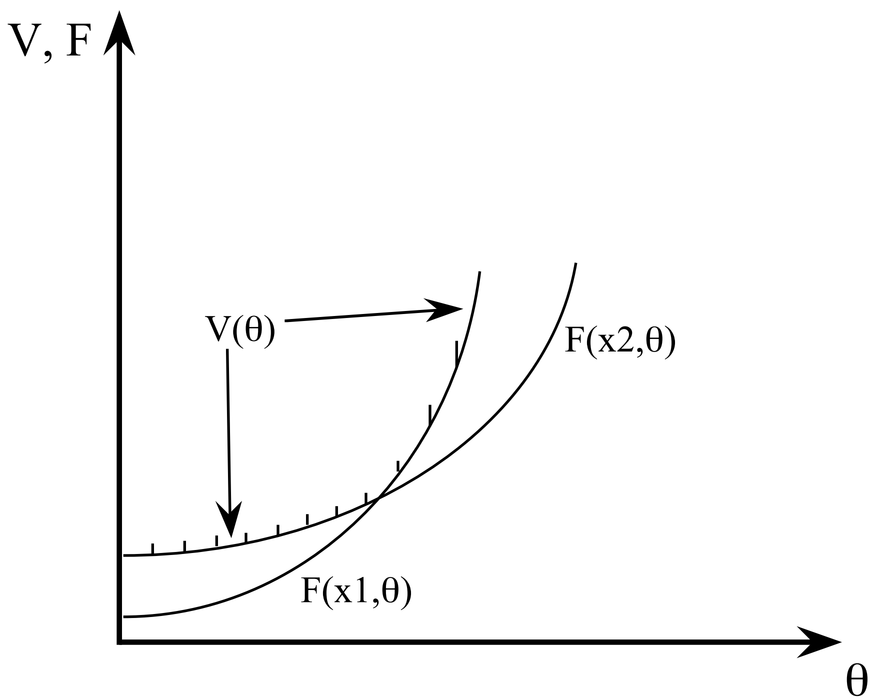
Example: $\pi(p, \alpha) = \max_{q \geq 0} \{\alpha p q - c(q)\}$, where $c(\cdot)$ is strictly increasing and strictly convex and $\alpha, p > 0$.

Since the objective function is affine in α, $\pi(p, \cdot)$ is convex in α. If the solution is unique, then, by the Envelope Theorem, $\partial\pi/\partial\alpha = pq^*(p, \alpha)$. But since π is convex in α, $\partial\pi/\partial\alpha$ is nondecreasing in α. And since $p > 0$, $q^*(p, \alpha)$ cannot decrease in α. (If π is C^2, we have

$$0 \leq \partial^2\pi/\partial\alpha^2 = p\partial q^*(p, \alpha)/\partial\alpha$$

so that $\partial q^*(p, \alpha)/\partial\alpha \geq 0$.)

The Maximum of a Collection of Convex Functions is Convex



$$V(\theta) = \max_{x}\{F(x,\theta)\}$$